

JKBOSE 2026

CLASS 11TH

RESPIRATION IN PLANTS

LECTURE-04

BOTANY

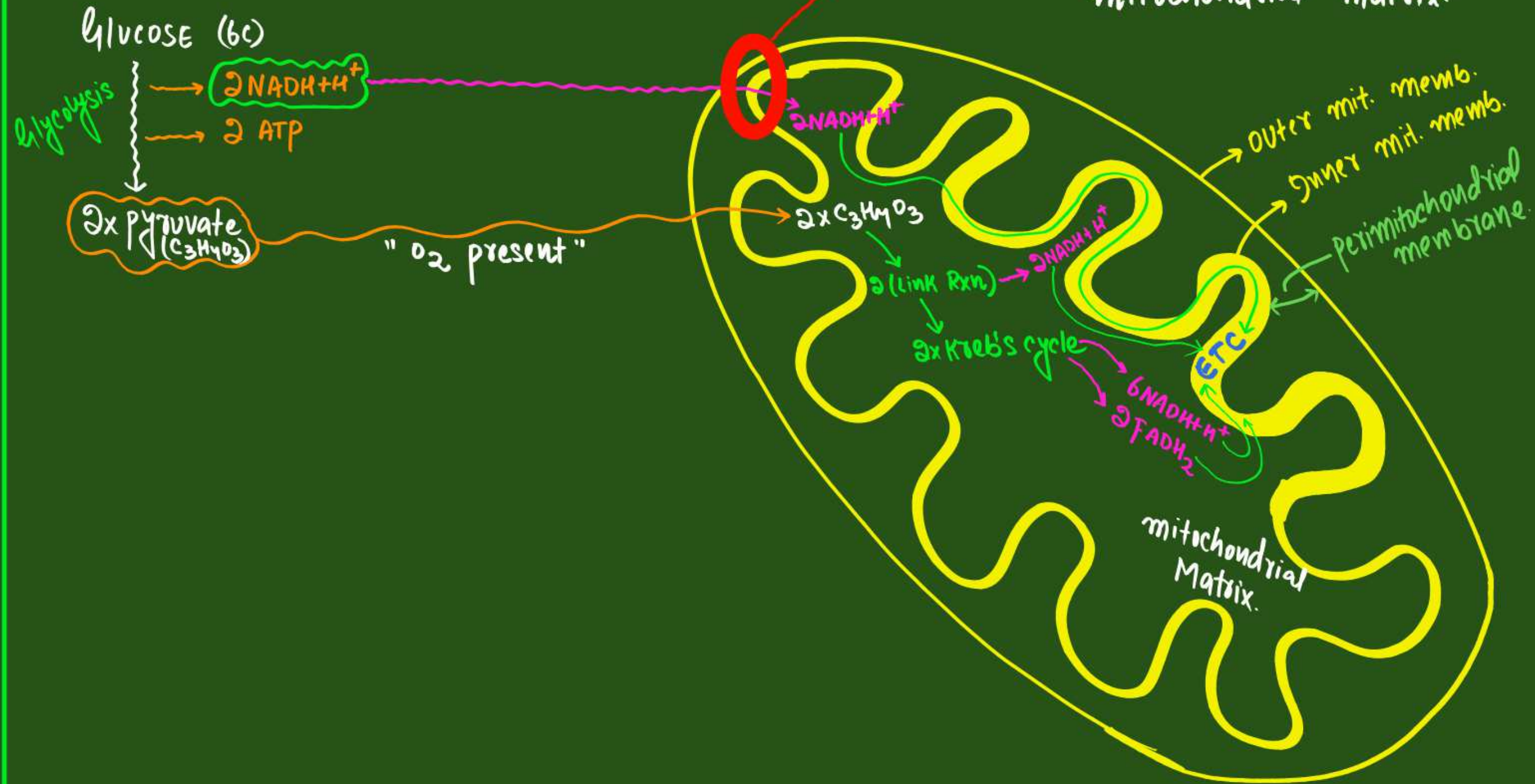
ARSHID BHAT SIR



Topics To Covered

- 1 Electron Transport Chain
 - 2 Oxidative Phosphorylation
 - 3 Amphibolic Pathway
 - 4 Respiratory Quotient
- Next lecture
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RECAP



ELECTRON TRANSPORT CHAIN

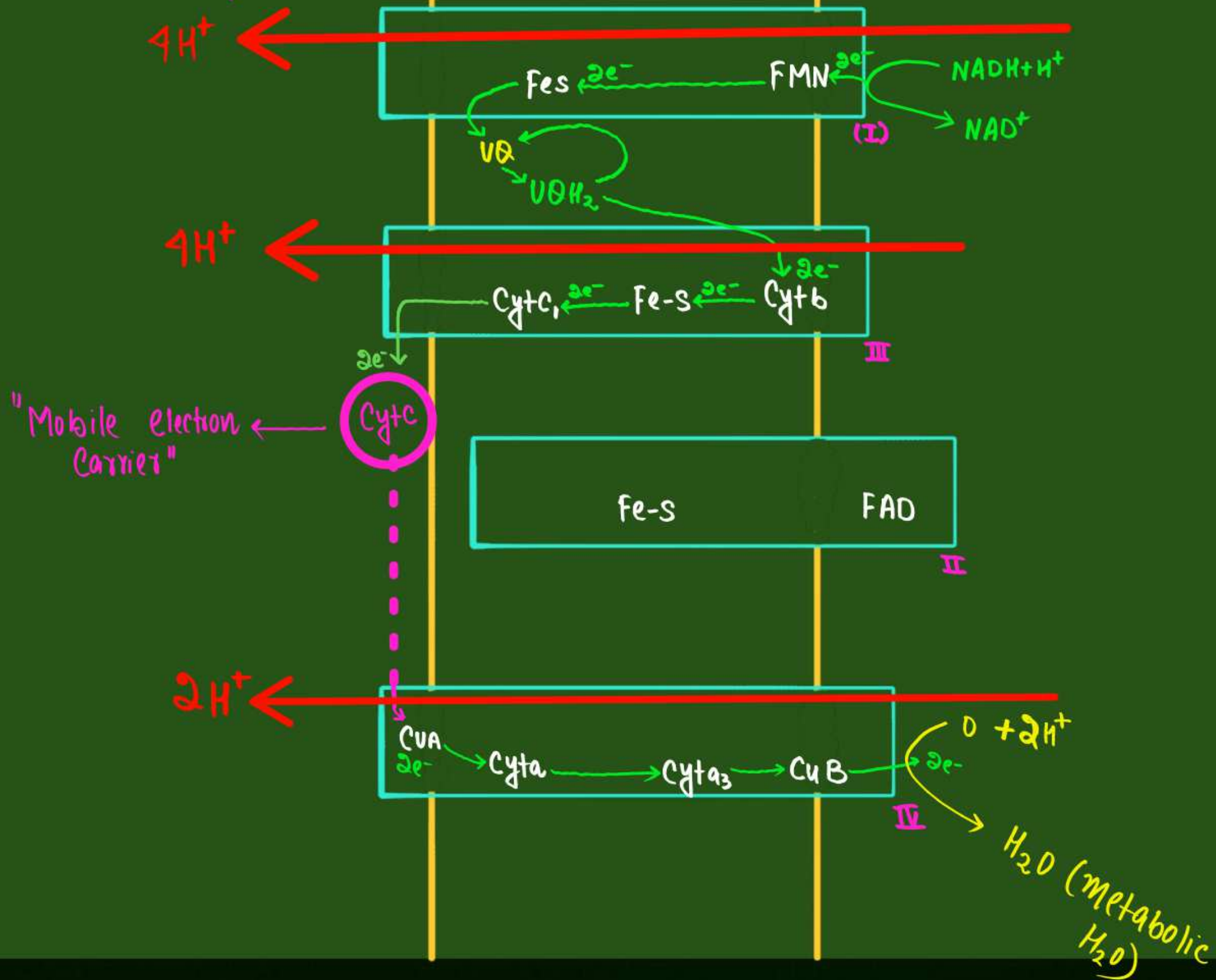
It involves transfer of electrons from $\text{NADH} + \text{H}^+$ / FADH_2 to O_2 via protein complexes.

Site of Electron transport chain (ETC) is "Inner mitochondrial membrane"

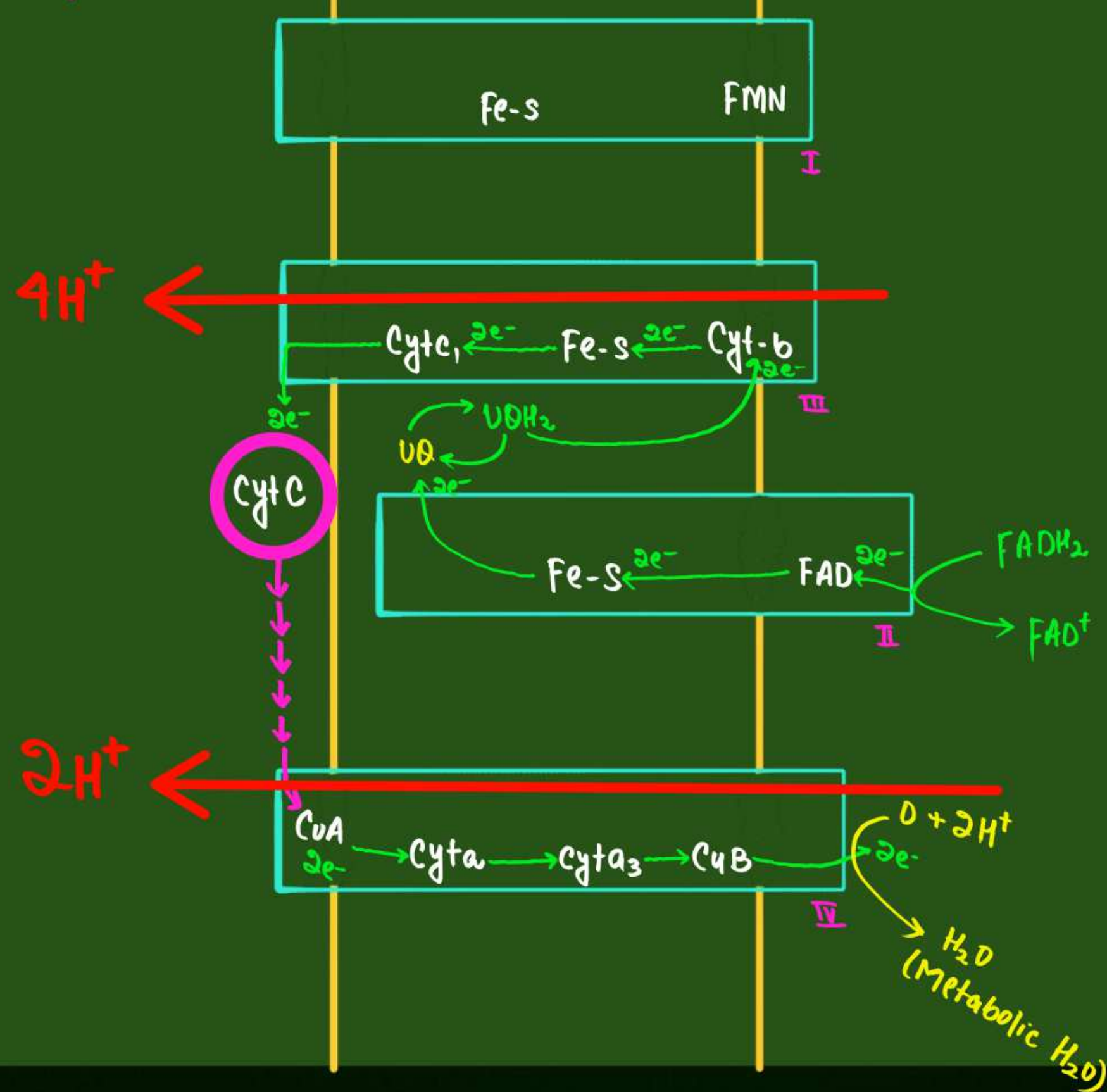
04 protein complexes involved in ETC

- ①. Complex - I (NADH dehydrogenase) $\longrightarrow$ Contains FMN and Fe-S
- ②. Complex - II (Succinate dehydrogenase) $\longrightarrow$ Contains FAD and Fe-S
- ③. Complex - III (Cyt-b_c complex) $\longrightarrow$ Contains Cyt b, Fe-S, Cyt c,
- ④. Complex - IV (Cyt-c oxidase) $\longrightarrow$ Contains CuA, Cyt a, Cyt a₃ & Cu-B.

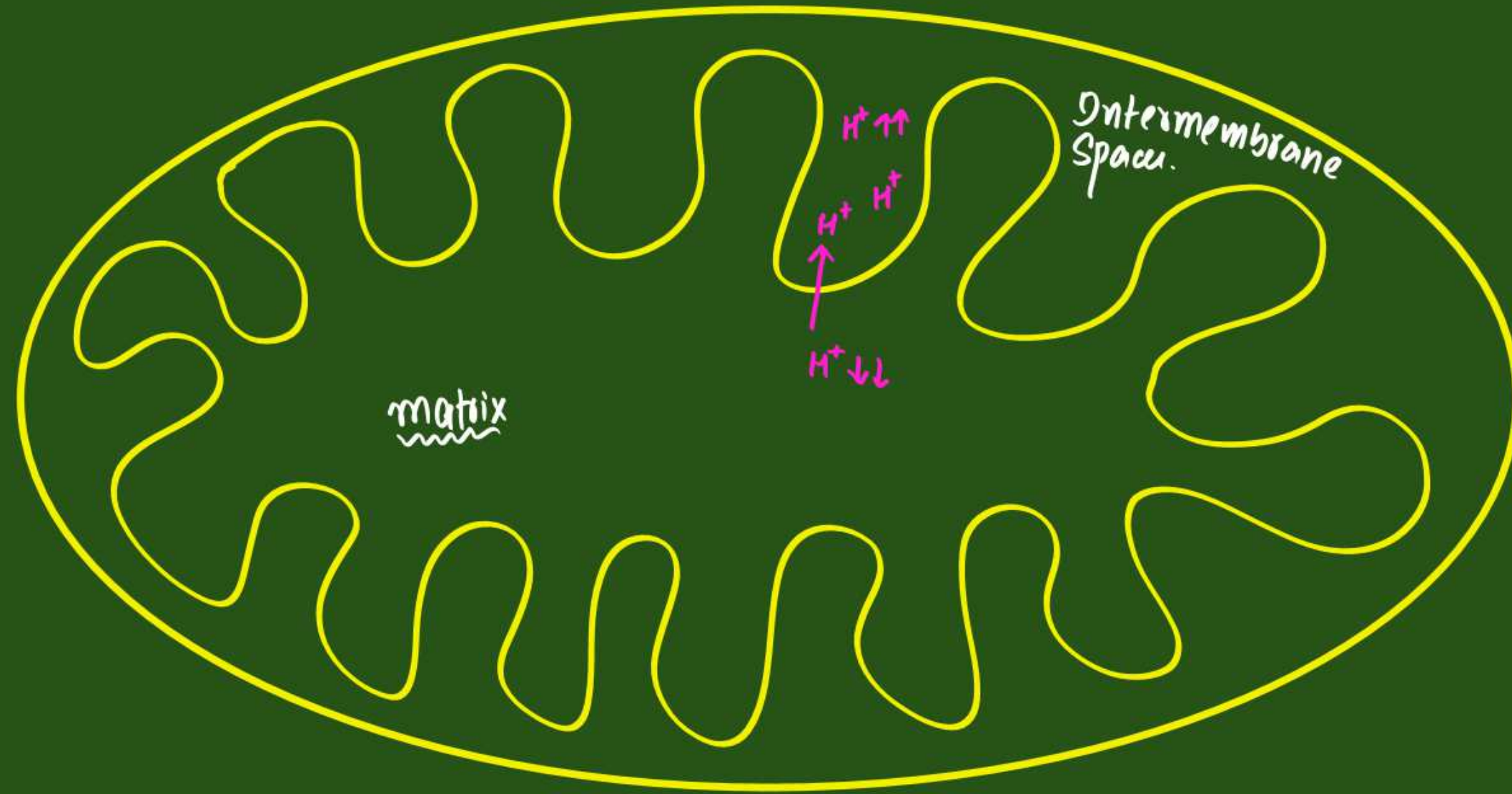
Inter membrane space ← I.M.M. → Mitochondrial matrix



Intermembrane Space I.M.M. Mitochondrial matrix

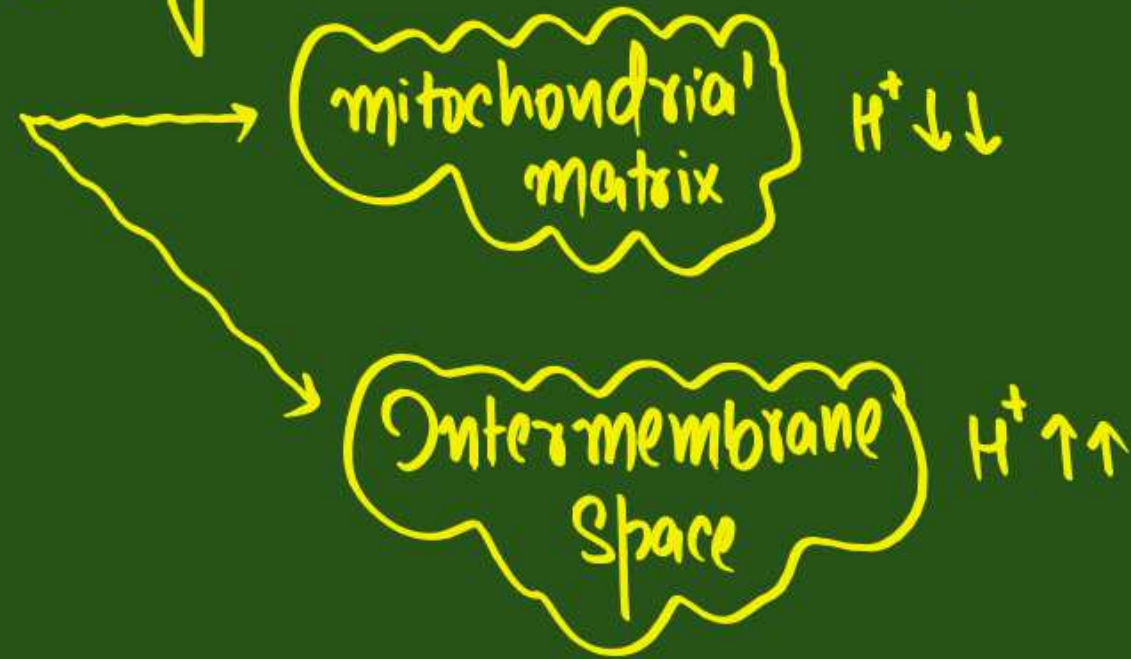


NADH + H⁺
FADH₂



Note:

1. Oxidation of 1 $\text{NADH} + \text{H}^+$ pumps 10 H^+ from mitochondrial matrix to intermembrane space.
2. Oxidation of 1 FADH_2 pumps 6 H^+ from mitochondrial matrix to intermembrane space.
3. Thus proton gradient is created across inner mitochondrial membrane.



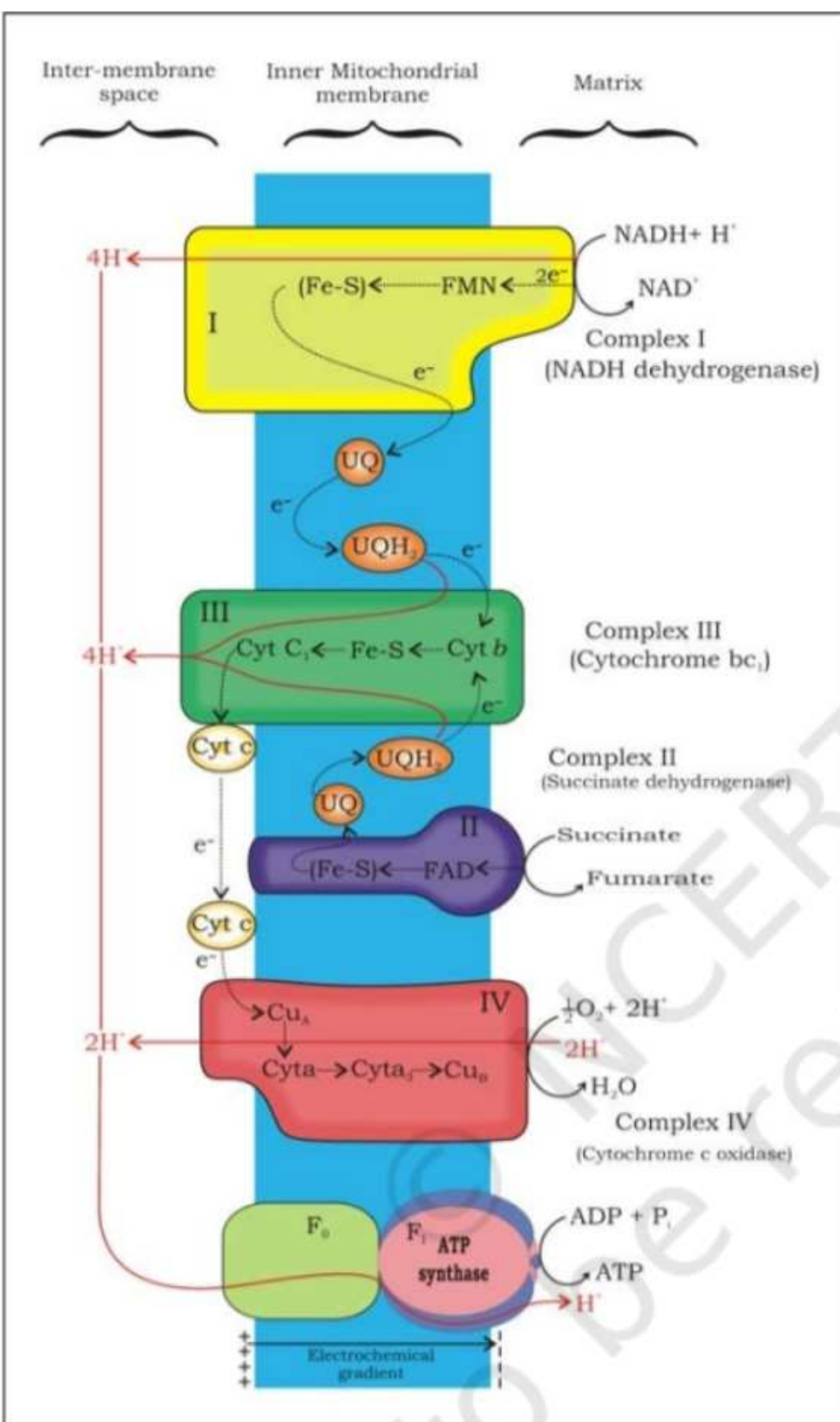
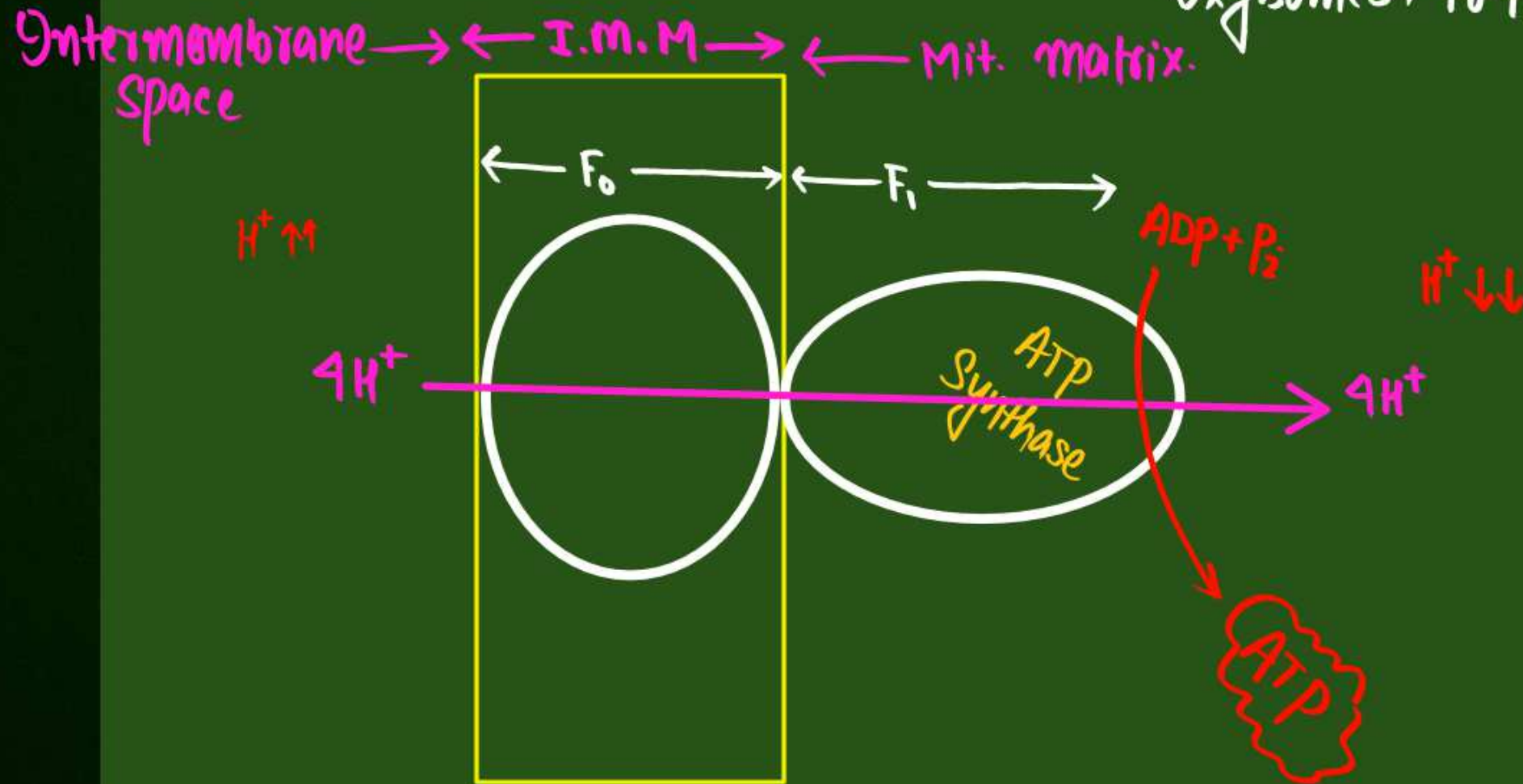


Figure 12.4 Electron Transport System (ETS)

OXIDATIVE PHOSPHORYLATION

≡ Formation of ATP from ADP and P_i , which is linked (coupled) to oxidation of $NADH+H^+$ & $FADH_2$.

≡ for oxidative phosphorylation → "Complex-V" is required
↓
Oxysomes / F_0-F_1 particle.



⊗ For formation of 1 ATP, 04 H^+ are being transferred from Intermembrane space to mitochondrial matrix through Complex-V.

∴ 1 NADH + H⁺ (oxidation) $\rightsquigarrow$ 10 H⁺ pump.

- 4 H⁺ = 1 ATP

- 10 H⁺ = 2.5 ATP $\approx$ 3 ATP

∴ 1 FADH₂ (oxidation) $\rightsquigarrow$ 6 H⁺ proton

- 4 H⁺ = 1 ATP

- 6 H⁺ = 1.5 ATP $\approx$ 2 ATP

QUESTIONS

Which of the following statements is incorrect?

[NEET-2021]

- A** Oxidation-reduction reactions produce proton gradient in respiration
- B** During aerobic respiration, role of oxygen is limited to the terminal stage
- ☒ **C** In ETC (Electron Transport Chain), one molecule of $\text{NADH} + \text{H}^+$ gives rise to 2 ATP molecules, and one FADH_2 gives rise to 3 ATP molecules
- D** ATP is synthesized through complex V

HOMEWORK

Revision V.V. import.
↓
gt is must...

Thank
You